TunFood: Food Recognition and Calorie Estimation using Deep Learning

Wiem Aouissaoui and Hanen Ameur

Isims, University of Sfax, Sfax, Tunisia

Introduction

Unbalanced nutrition has become a significant issue in our community, leading to serious illnesses due to unhealthy lifestyles. To address this, it is crucial to manage our caloric intake effectively.

- Developing a system for food recognition and calorie estimation could provide a practical solution.
- However, there is a lack of research on Tunisian food for both recognition and caloric estimation tasks.

Objectives

Our main goals for this project are:

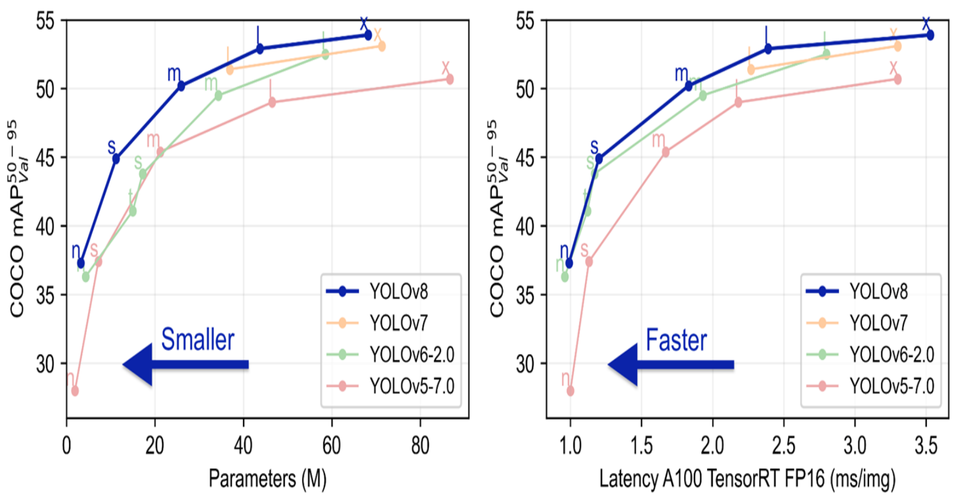
- Develop a web app capable of identifying a variety of foods, including Tunisian dishes, from a simple photo taken by the user.
- Enable the web app to estimate the amount of calories in Tunisian food.
- Provide guidance to tourists, allowing them to explore traditional Tunisian foods.

Contributions

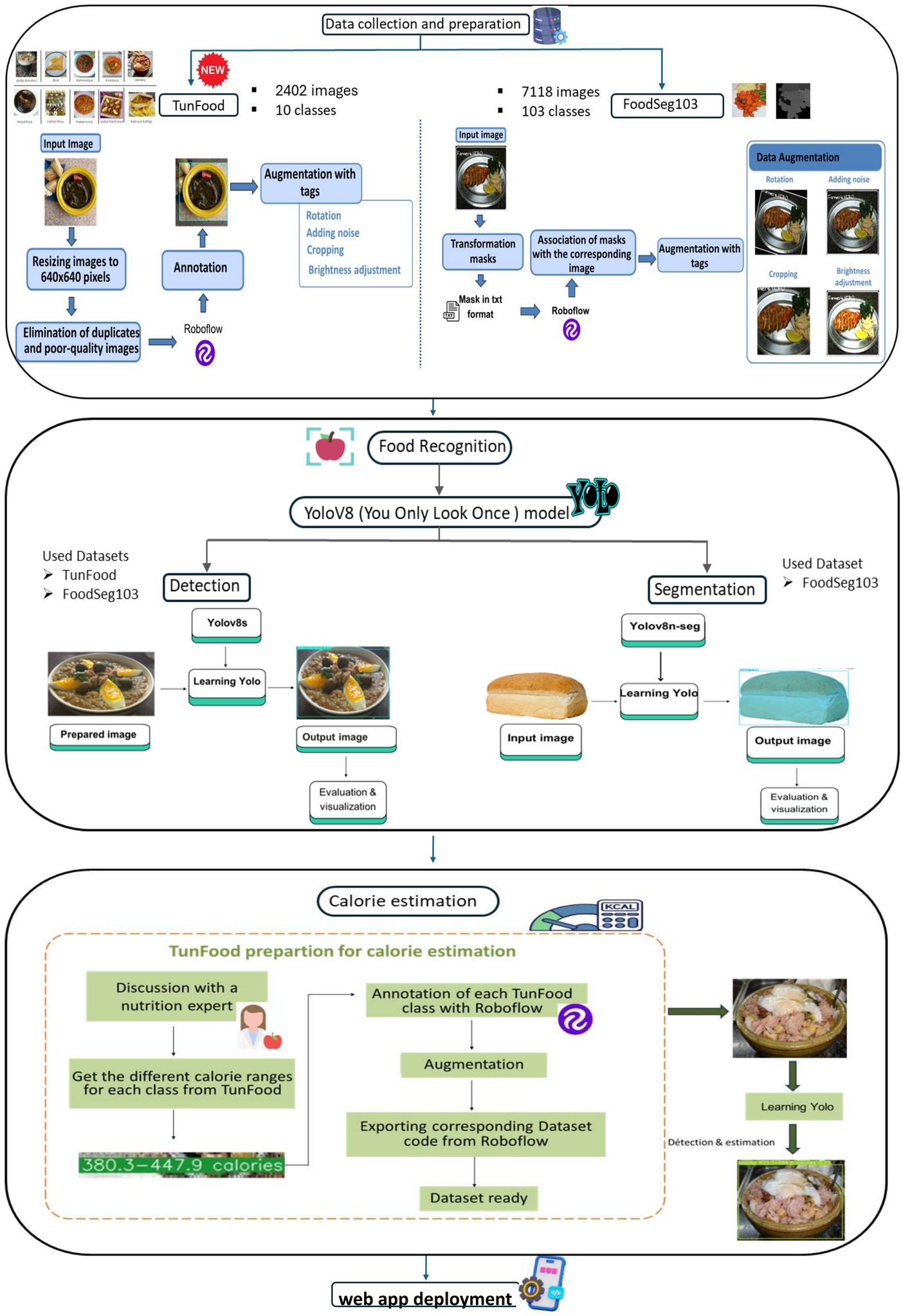
- Applying a novel method for segmentation and detection using YOLOv8.
- A newly collected dataset for Tunisian food.
- Achieving state-of-the-art performance on the FoodSeg103 dataset, outperforming existing methods, as well as promising results on TunFood.

Yolov8

- YOLOv8 (You Only Look Once) is a real-time object detection model known for its advanced capabilities in object detection, segmentation, and classification.
- Developed by Ultralytics on January 10, 2023.
- A state-of-the-art model in its field, renowned for its success and superior performance compared to previous versions.



Proposed method



Experimental results

Segmentation

- Comparison with state-of-the-art segmentation models using FoodSeg103

Method	mIou(%)	aAcc(%)	mAcc (%)
FPN	27.28	75.23	36.7
CCNet	28.6	78.9	47.8
Deeplabv3+	34.2	-	-
Upernet	39.2	82.02	52.27
InternImage-B	41.1	-	-
BEiT v2	49.4	-	-
FoodSAM	46.42	84.10	58.27
YOLOV8(our model)	71.2	87.1	85.3

- Results from the FoodSeg103 dataset for segmentation

Model	Precision(%)	Accuracy(%)	Recall(%)	F1-score(%)
Yolov8	97	81.94	72.96	82.03

Detection results

- Comparison with state-of-the-art models using TunFood

Model	mAP(0.5)(%)	mAP(0.5-0.95)(%)	Precision(%)	Recall(%)
Yolov4	90.9	56,7	70.6	90
Yolov4	93.8	62.5	89.9	89.8
Yolov8	91.9	61.1	91.8	88

Calorie estimation results

- Calorie estimation results using TunFood

Model	mAP(0.5)(%)	mAP(0.5-0.95)(%)	Precision(%)	Recall(%)
Yolov8	94	65.02	90	90.2

Conclusion and Future Work

- Transition the system into a mobile app for greater convenience.
- Improve the accuracy and precision of calorie estimation, ensuring exact calculations.
- Expand the dataset to include a wider variety of foods, enhancing the app's efficiency.

Contacts

- Aouissaoui wiem: wiemaouissaoui26@gmail.com
- Ameur Hanen: hanene.ameur@isims.usf.tn